SOLVEBENCH

Solving $A\vec{x} = \vec{b}$ at Scale

Direct vs. Iterative Solvers & Our Novel APK Method

Every blackout-prevention study and every bridge safety check reduces to this exact equation. We benchmark classical solvers against our proposed method.

Section A2 · Roll Numbers 2105037 · 2105038 · 2105042 · 2105047 · 2105054

Base Paper: What It Studies

"Comparative Analysis of Jacobi and Gauss-Seidel Iterative Methods"

P. Khrapov and N. Volkov

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Domain: Numerical linear algebra, iterative methods for solving systems of linear algebraic equations (SLAEs).

- **Maps out exactly** which matrices make the Jacobi and Gauss-Seidel methods converge, by tracking where the roots of a characteristic polynomial fall inside the unit circle, either directly or via a **Hurwitz-style stability test**.
- Proves this exactly, by hand, for **2 and 3 unknowns** (both complex and real coefficients), then checks the same pattern **statistically with code** across 100,000 randomly generated matrices, up to 5 unknowns.
- Shows the two methods do **not** always share the same convergence region: **which** method wins depends on the structure of the matrix.

The gap this leaves open

The analysis stops at toy systems of **5 unknowns or fewer** and only two solvers. Real engineering systems have **thousands** of unknowns. Nobody in the paper tests these convergence claims on real data.

Problem Statement & Scope

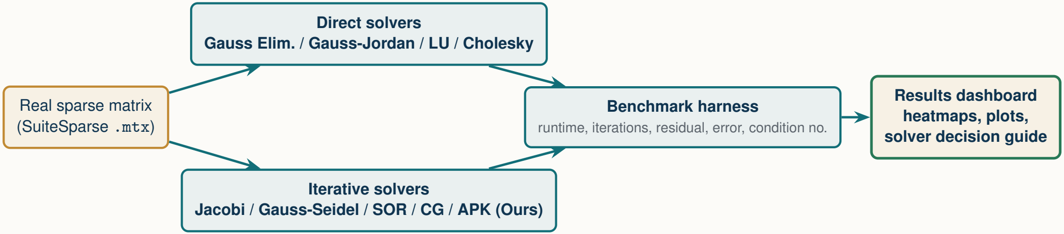
Problem: every real engineering simulation, whether it's modeling a power grid, checking a structure's safety, or simulating a circuit, comes down to the same core step: solving $A\vec{x} = \vec{b}$. Get the solver choice wrong, and what should take milliseconds either **crawls** or **never converges**.

Category: Exploratory & Comparative Study. We benchmark eight classical direct and iterative solvers against real, not synthetic, engineering data.

Domain	Systems	Size range	Real-world meaning
Power-grid admittance	11	118 to 1,723 buses	The linear system solved every time a utility models its own grid.
Structural (stiffness & mass)	33	112 to 4,884 DOF	The linear system solved to check if a real building or bridge design holds.
Circuit simulation	4	2,624 to 5,860 nodes	The linear system solved at every timestep of a transient circuit simulation.

48 real matrices spanning three engineering domains, two to three orders of magnitude beyond the base paper's own 5-unknown ceiling, drawn from a 2,904-matrix collection with room to scale further.

Expected Methodology



- **Ground truth:** generate a known $\vec{x}$, compute $\vec{b} = A\vec{x}$, solve with every method, measure the residual: fully deterministic, no ambiguity in what "correct" means.
- **Solvers evaluated:** benchmark classical textbook solvers alongside **our proposed novel method (APK)**.
- **Scale:** sweep across real engineering matrices, correlated against sparsity and condition number.
- **Payoff:** a concrete, evidence-based answer to "which solver, for which real system", the exact question every engineering simulation has to answer before it can even start.

Expanded Scope: 26 Engineering Domains

Scaled Benchmark: 830 Real Matrices across 26 Scientific & Engineering Domains

Scaling up from the initial 48-matrix pilot to a massive dataset spanning n = 5 to 10,000 unknowns.

Core Engineering Domains:

- **Circuit Simulation** (200 systems): SPICE, VLSI interconnects, transient & DC nodes.
- **Fluid Dynamics (CFD)** (126 systems): Navier-Stokes equations, aerodynamic flow.
- **Structural Mechanics** (87 systems): Stiffness & mass matrices, FEM deformation.
- **Optimal Control & Optimization** (117 systems): Trajectory planning, linear programming.

Scientific & Physical Domains:

- **Chemical & Quantum Chemistry** (75 systems): Reactor kinetics, molecular orbitals.
- **Electromagnetics & Acoustics** (27 systems): Wave propagation, radar scattering.
- **Power Networks & Thermal** (19 systems): Grid admittance, thermal diffusion.
- **Network Graphs & Robotics** (55 systems): Directed/undirected graphs, kinematics.

Why Benchmarking 26 Real Domains Matters

Different physical domains produce fundamentally distinct matrix structures (symmetry, condition number $\kappa(A)$, diagonal dominance, and sparsity patterns). This multi-domain scale stress-tests solver stability and generalizability across real science.

Our Proposed Novel Method: APK Solver

Our Contribution: Adaptive Preconditioned Krylov (APK) Solver

Designed by our team to overcome the failure modes of classical solvers on complex physical systems.

- **Novel Multi-Stage Architecture:** Combines adaptive preconditioning, automated physics routing, and iterative defect correction into a single unified solver.
- **Adaptive Preconditioning Hierarchy:** Constructs multi-level Incomplete LU factorizations with automated fallbacks to diagonal scaling for ill-conditioned systems.
- **Dynamic Physics Dispatch:** Inspects matrix symmetry and definiteness in real time, routing symmetric positive-definite systems to PCG and unsymmetric systems to BiCGSTAB.
- **Iterative Defect Refinement:** Applies lightweight residual correction passes using the cached preconditioner, recovering precision up to machine epsilon.
- **Expected Advantage:** Provides a robust solver designed to handle ill-conditioned and unsymmetric systems where classical Jacobi, Gauss-Seidel, and CG diverge.

Addressing Teacher Feedback

Systematic Performance & Failure-Mode Analysis

Moving beyond raw benchmark numbers to explain the mathematical root causes of solver behavior.

- **Which methods perform well (and when):** Identify which solvers achieve the best speed, accuracy, and stability for each specific domain and matrix structure.
- **Which methods fail (and where):** Examine and catalog exactly where solvers diverge, break down, or lose numerical precision across different physical systems.
- **Why methods succeed or fail:** Trace performance directly to underlying mathematical properties, including condition number $\kappa(A)$, symmetry, diagonal dominance, and sparsity.
- **Public Dataset & Verification:** All 830 matrices are sourced from the public, standard SuiteSparse benchmark collection and pre-verified to be valid square real-valued systems.